

DBMS SQL Queries – Syntax and Examples

CREATE TABLE

```
CREATE TABLE table_name (  
column1 datatype [constraint],  
column2 datatype [constraint],  
...  
);
```

Example:

```
CREATE TABLE Employees (  
emp_id INT PRIMARY KEY,  
emp_name VARCHAR(50),  
age INT,  
salary DECIMAL(10,2),  
department VARCHAR(30)  
);
```

ALTER TABLE

```
ALTER TABLE table_name ADD column_name datatype;  
ALTER TABLE table_name MODIFY column_name new_datatype;  
ALTER TABLE table_name DROP COLUMN column_name;
```

Example:

```
ALTER TABLE Employees ADD date_of_join DATE;
```

INSERT

```
INSERT INTO table_name (column1, column2, ...) VALUES (value1, value2, ...);
```

Example:

```
INSERT INTO Employees (emp_id, emp_name, age, salary, department)  
VALUES (101, 'Rahul Kumar', 25, 35000.00, 'IT');
```

DELETE

```
DELETE FROM table_name WHERE condition;
```

Example:

```
DELETE FROM Employees WHERE emp_id = 101;
```

UPDATE

```
UPDATE table_name  
SET column1 = value1, column2 = value2  
WHERE condition;
```

Example:

```
UPDATE Employees
```

```
SET salary = 40000.00  
WHERE emp_id = 102;
```

DROP TABLE

```
DROP TABLE table_name;
```

Example:

```
DROP TABLE Employees;
```

GROUP BY

```
SELECT column_name, aggregate_function(column_name)  
FROM table_name  
GROUP BY column_name;
```

Example:

```
SELECT department, AVG(salary) AS Avg_Salary  
FROM Employees  
GROUP BY department;
```

ORDER BY

```
SELECT column1, column2  
FROM table_name  
ORDER BY column_name [ASC|DESC];
```

Example:

```
SELECT emp_name, salary  
FROM Employees  
ORDER BY salary DESC;
```

INNER JOIN

```
SELECT table1.column, table2.column  
FROM table1  
INNER JOIN table2  
ON table1.common_column = table2.common_column;
```

Example:

```
SELECT e.emp_name, d.dept_name  
FROM Employees e  
INNER JOIN Departments d  
ON e.department = d.dept_id;
```

LEFT OUTER JOIN

```
SELECT table1.column, table2.column  
FROM table1  
LEFT JOIN table2  
ON table1.common_column = table2.common_column;
```

Example:

```
SELECT e.emp_name, d.dept_name
FROM Employees e
LEFT JOIN Departments d
ON e.department = d.dept_id;
```

RIGHT OUTER JOIN

```
SELECT table1.column, table2.column
FROM table1
RIGHT JOIN table2
ON table1.common_column = table2.common_column;
```

Example:

```
SELECT e.emp_name, d.dept_name
FROM Employees e
RIGHT JOIN Departments d
ON e.department = d.dept_id;
```

FULL JOIN

```
SELECT * FROM table1
LEFT JOIN table2 ON table1.id = table2.id
UNION
SELECT * FROM table1
RIGHT JOIN table2 ON table1.id = table2.id;
```

Example:

```
SELECT e.emp_name, d.dept_name
FROM Employees e
LEFT JOIN Departments d ON e.department = d.dept_id
UNION
SELECT e.emp_name, d.dept_name
FROM Employees e
RIGHT JOIN Departments d ON e.department = d.dept_id;
```

Aggregate Functions

COUNT() - Counts number of rows
SUM() - Returns total sum
AVG() - Returns average value
MAX() - Returns maximum value
MIN() - Returns minimum value

Example:

```
SELECT COUNT(*) FROM Employees;
SELECT AVG(salary) FROM Employees;
```

Stored Procedure – INSERT

```
DELIMITER //
```

```
CREATE PROCEDURE InsertEmployee(  
  IN id INT,  
  IN name VARCHAR(50),  
  IN age INT,  
  IN salary DECIMAL(10,2),  
  IN dept VARCHAR(30)  
)  
BEGIN  
  INSERT INTO Employees VALUES (id, name, age, salary, dept);  
END //
```

```
DELIMITER ;
```

Example Call:

```
CALL InsertEmployee(106, 'Neha Singh', 27, 42000.00, 'Finance');
```

Stored Procedure – UPDATE

```
DELIMITER //
```

```
CREATE PROCEDURE UpdateSalary(  
  IN empid INT,  
  IN new_salary DECIMAL(10,2)  
)  
BEGIN  
  UPDATE Employees  
  SET salary = new_salary  
  WHERE emp_id = empid;  
END //
```

```
DELIMITER ;
```

Example Call:

```
CALL UpdateSalary(103, 50000.00);
```

Stored Procedure – DELETE

```
DELIMITER //
```

```
CREATE PROCEDURE DeleteEmployee(IN empid INT)  
BEGIN  
  DELETE FROM Employees WHERE emp_id = empid;  
END //
```

```
DELIMITER ;
```

Example Call:

```
CALL DeleteEmployee(105);
```